

```
In [1]: # Behzad Ghabaei
# CS 82A - Data Science
# module 4 Lab 1 - Describe and Correlate
# module4_lab.ipynb
# Instructor Seno
# 9/22/2026
```

Step 1: Load your clean data

Load `sales_clean.csv` and confirm its shape matches what you saved in Lab 3. You should see: (292, 6)

```
In [2]: import pandas as pd
df = pd.read_csv("sales_clean.csv")
df.shape
```

```
Out[2]: (292, 6)
```

Step 2: Center and spread of price and qty

Compute mean, median, and standard deviation for price and for qty (Tutorial 1 pattern). You should see: six numbers. The price mean sits noticeably above the price median; that gap is seeded skew.

```
In [3]: print("price mean: ", df["price"].mean())
print("price median: ", df["price"].median())
print("price std: ", df["price"].std())
print("-" * 10)
print("qty mean: ", df["qty"].mean())
print("qty median: ", df["qty"].median())
print("qty std: ", df["qty"].std())
```

```
price mean:  55.52315068493151
price median:  37.53
price std:  71.76708493504132
-----
qty mean:  2.9863013698630136
qty median:  3.0
qty std:  1.4092785381929205
```

Step 3: Which summary would you report? (markdown cell)

Which of mean or median would you report for price, and why? 2–3 sentences.

Remembering our Cleaning Log from `module3_lab2` recall imputing for price values at 37.53. Since the price standard deviation is such a large amount, at 71.76, I would prefer the median, and inspect the mean for outstanding outliers. For example 479.29 is a large sum from "price" listed as the max and which is pulling the mean up, this is for expensive electronic equipment in quantity. When we take a look at the prices that carry weight, we notice that the merchandise is a mixture of very different items of value, for example, we're comparing the prices of pen and paper, vs. cameras and monitors, so there is going to be a significant difference, but these outliers are really important for inventory reference and loss prevention. It's important to keep the highly priced items accounted for, but lower priced items are not that important, since these things are pens, paper, and mugs. The min at 2.93 is listed as 4 notebooks, but we used the median at 37.53 to fill empty values, which explains the single notebook for 37.53, which is high. So rather than filling in the missing values by hand, these imputed price values were given 37.53.

```
In [4]: df.describe()
```

```
Out[4]:
```

	order_id	price	qty	zip
count	292.000000	292.000000	292.000000	292.000000
mean	1145.500000	55.523151	2.986301	45463.989726
std	84.437354	71.767085	1.409279	39284.386854
min	1000.000000	2.930000	1.000000	2116.000000
25%	1072.750000	11.045000	2.000000	2134.000000
50%	1145.500000	37.530000	3.000000	30303.000000
75%	1218.250000	55.510000	4.000000	90210.000000
max	1291.000000	479.290000	5.000000	98101.000000

These prices were imputed.

```
In [5]: df[df["price"] == 37.53]
```

```
Out[5]:
```

	order_id	date	product	price	qty	zip
3	1066	2026-05-30	notebook	37.53	1	98101
5	1280	2026-04-22	pen set	37.53	4	30303
30	1088	2026-04-28	pen set	37.53	2	2134
44	1101	2026-05-12	keyboard	37.53	3	2116
49	1199	2026-04-18	charger	37.53	5	30303
50	1018	2026-05-04	monitor	37.53	4	10001
55	1170	2026-04-04	monitor	37.53	1	2116
91	1040	2026-05-08	pen set	37.53	3	2134
103	1240	2026-04-16	headphones	37.53	1	60614
127	1261	2026-05-09	desk lamp	37.53	2	90405
176	1005	2026-05-24	monitor	37.53	2	2134
245	1155	2026-04-08	charger	37.53	3	2134

This was the max expensive priced item, a computer monitor, quantity 1.

```
In [6]: df[df["price"] == 479.29]
```

```
Out[6]:
```

	order_id	date	product	price	qty	zip
221	1093	2026-04-14	monitor	479.29	1	90210

This was the min priced item, notebook, quantity 4.

```
In [7]: df[df["price"] == 2.93 ]
```

```
Out[7]:
```

	order_id	date	product	price	qty	zip
89	1069	2026-04-18	notebook	2.93	4	60614

```
In [8]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 292 entries, 0 to 291
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype  
---  -
0   order_id    292 non-null    int64  
1   date        292 non-null    str     
2   product     292 non-null    str     
3   price       292 non-null    float64 
4   qty         292 non-null    int64  
5   zip         292 non-null    int64  
dtypes: float64(1), int64(3), str(2)
memory usage: 13.8 KB
```

```
In [9]: df.shape
```

```
Out[9]: (292, 6)
```

```
In [10]: df.head()
```

```
Out[10]:
```

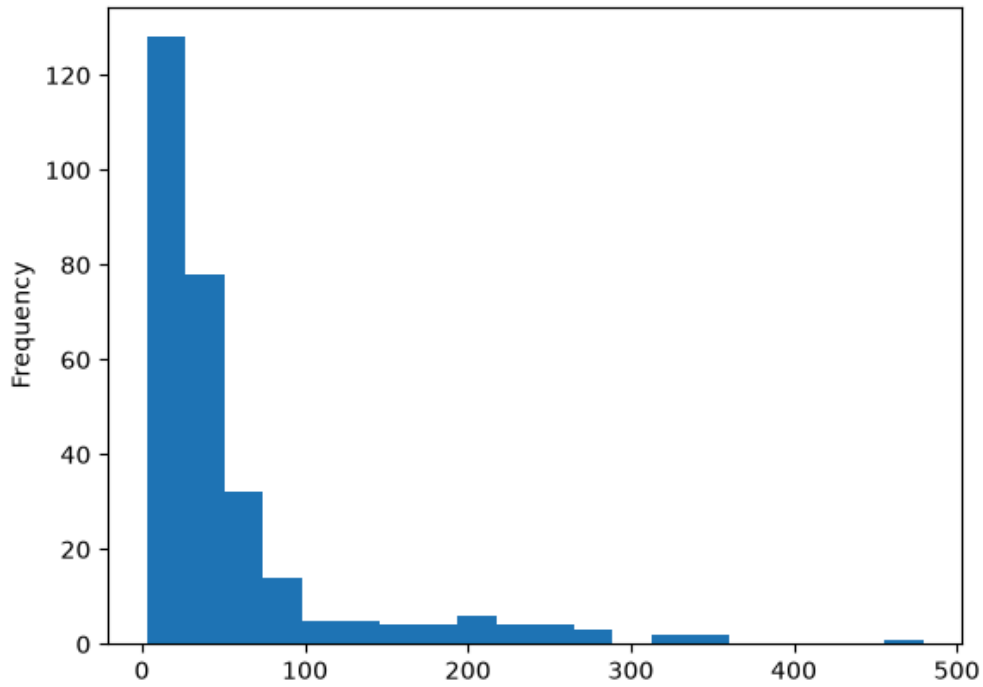
	order_id	date	product	price	qty	zip
0	1254	2026-04-06	mug	11.36	4	60614
1	1057	2026-05-30	charger	14.79	3	30303
2	1150	2026-05-06	mug	5.50	1	10001
3	1066	2026-05-30	notebook	37.53	1	98101
4	1114	2026-04-06	webcam	45.59	4	90405

Step 4: Histogram of price

Plot a histogram of price with about 20 bins (Tutorial 2) and, in a markdown line, name its shape. You should see: a right-skewed histogram: most prices low, a tail of expensive items.

```
In [11]: df["price"].plot(kind="hist", bins=20)
```

```
Out[11]: <Axes: ylabel='Frequency'>
```



The histogram shows a skewed right tail, with the bulk on the left, this signifies the mean is above the median. The lack of symmetry, is called a positive skew. The skew favors the median. Expensive items have low frequency price around 500, and inexpensive items have high frequency, less than 100 frequently appears.

Step 5: Create the revenue column

Make revenue = price * qty as a new column (Tutorial 3 pattern). You should see: a 7th column appears in `df.head()`.

The revenue column is useful to see how the quantity can change the actual revenue due to the price.

```
In [12]: df["revenue"] = df["price"] * df["qty"]
df.head()
```

```
Out[12]:
```

	order_id	date	product	price	qty	zip	revenue
0	1254	2026-04-06	mug	11.36	4	60614	45.44
1	1057	2026-05-30	charger	14.79	3	30303	44.37
2	1150	2026-05-06	mug	5.50	1	10001	5.50
3	1066	2026-05-30	notebook	37.53	1	98101	37.53
4	1114	2026-04-06	webcam	45.59	4	90405	182.36

Step 6: Correlation matrix

Compute the correlation matrix of the numeric columns (Tutorial 4). Note the strongest off-diagonal value. You should see: a matrix; qty vs. revenue should be the standout.

```
In [13]: df.corr(numeric_only=True)
```

```
Out[13]:
```

	order_id	price	qty	zip	revenue
order_id	1.000000	-0.005070	0.031420	-0.014784	0.004181
price	-0.005070	1.000000	-0.041982	0.023812	0.821001
qty	0.031420	-0.041982	1.000000	0.086869	0.312793
zip	-0.014784	0.023812	0.086869	1.000000	0.047090
revenue	0.004181	0.821001	0.312793	0.047090	1.000000

Pearson's r , measures how closely two variables follow a straight line relationship. It always falls between -1 and +1. Positive means they rise together, and negative means, as one rises the other falls. Zero means no relationship at all. The values near 1 or -1 mean that they are close to a straight line. Values near 0 form a shapeless cloud. For example, 0.1 is weak, 0.3 is moderate, and above 0.5 is strong (negative also). If there was a curved perfect relationship, the r could be very low. Correlations describe that changes in one variable are associated by another. However with correlations just because 2 things appear to have a strong linear relationship, doesn't mean they are related. Such as spurious correlations, that appear to be related but are not directly, but a regression is a functional relationship. Predictable analytics that provide actionable insights require logical and accurate datasets. From our lab, a correlation of 0.31 is a weak positive link, between qty and revenue, and a strong positive relationship between price and revenue is $r = .82$. The other values are not really related to economics, like zip and id, and indicate no relationship. There seems to be a strong linear relationship between **price increase and revenue production**, an even higher correlation than between **quantity and revenue**, which is at .31, a moderate or even low, poorly related correlation.

-.005 indicates no relationship

.031 indicates no relationship

-.0149 indicates no relationship

.0041 indicates no relationship

.086 indicates no relationship

.312 indicates weak or moderate relationship

-.014 indicates no relationship

.023 indicates no relationship

.047 indicates no relationship

.821 indicates **strong** linear positive relationship. (remarkable)

```
In [14]: df["qty"].corr(df["revenue"])
```

```
Out[14]: np.float64(0.3127930967032907)
```

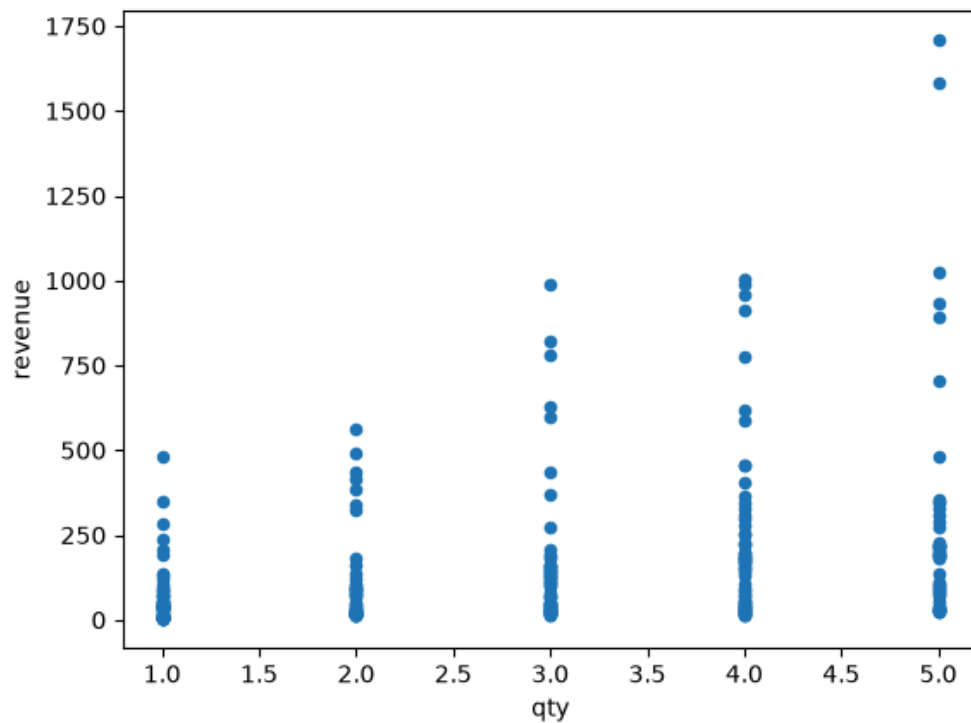
I noticed the diagonal value at 1.0 and the qty and revenue correlation at .3127, which is a weak positive linear relationship, The strong relationship is between price and revenue.

Step 7: Scatter to verify

Scatter-plot qty against revenue (Tutorial 5) and check the cloud agrees with the r value. You should see: an upward-drifting cloud.

```
In [15]: df.plot.scatter(x="qty", y="revenue")
```

```
Out[15]: <Axes: xlabel='qty', ylabel='revenue'>
```



The upward drifting cloud represents the revenue increasing as the quantity increases slightly. It verifies that there is a weak relationship between revenue and quantity.

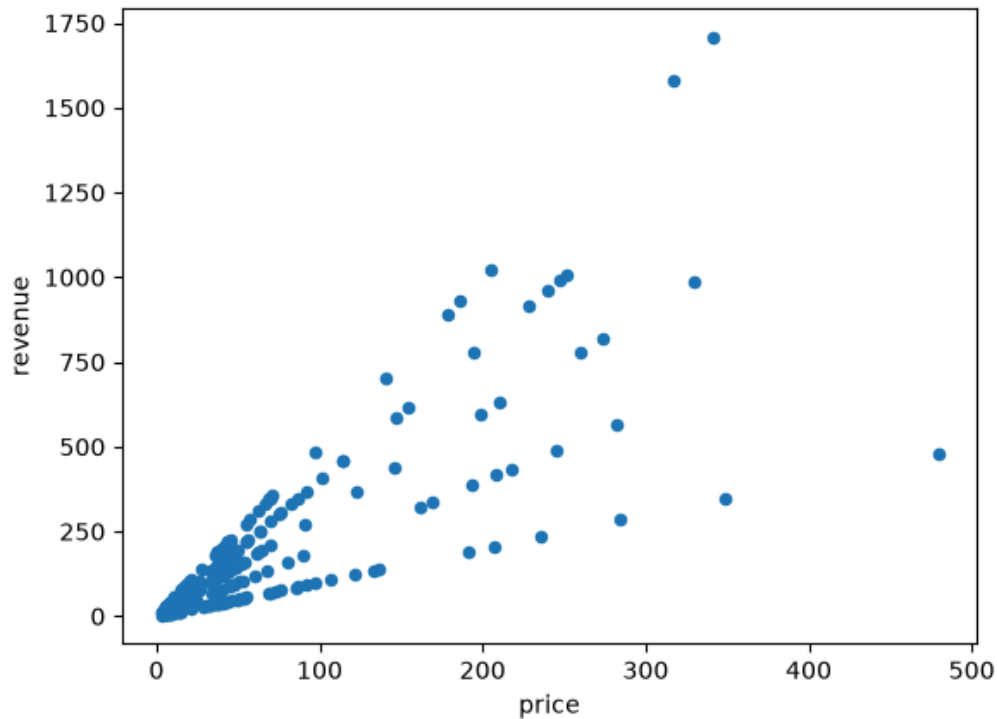
By comparison the correlation of price and revenue is strong at .82 , scatter plot indicates below.

```
In [16]: df["price"].corr(df["revenue"])
```

```
Out[16]: np.float64(0.8210012800930012)
```

```
In [17]: df.plot.scatter(x="price", y="revenue")
```

```
Out[17]: <Axes: xlabel='price', ylabel='revenue'>
```



Step 8: The analyst's sentence (markdown cell)

One plain-English claim: the relationship, its strength, and one caution (a confounder or outlier risk). This is the deliverable that matters most.

The conclusion is that expensive items are the reason behind large revenue results. There is a weak positive correlation between qty and revenue. The r describes the relationship between the two. **The $r = .3127$, between qty and revenue is low**, but a remarkable correlation occurs between **price and revenue the $r = .82$** , considerably higher and better. The correlations with very small values signify that the two variables don't work together, such as qty and price. Negative values approaching -1 are also a strong correlation, for example -0.9 indicates strong negative correlation.

These correlation values may reflect that care is not taken, by making cheap priced items, purchased in bulk and expensive purchased items purchased one at a time. The other values like zip and id were not as helpful to use as price, quantity, and revenue. What to say to the manager is that the data shows a **strong positive relationship between price and revenue ($r = 0.821$)**, this means that total revenue is related to item price but not quantity, and there are a few highly priced outliers that may skew the metrics.

Another point is that if the 12 items could have been filled individually by hand, would this result in more accurate correlations? If the values could have been inserted differently with more careful prices, the correlation between qty and revenue would change, for example to $r = .2785$. But the correlation between price and revenue would be around at $r = .8688$. This could show us that accurate insertions would have gotten different results with a slightly higher correlation between price and revenue. Imputing with the median is a good way of balancing the dataset with empty data points, without having to fill each missing price individually. It seems convenient and accurate when comparing .82 vs .86.

Here is the hand insertion sample

```
df.loc[3, "price"] = 6.00
```

```
df.loc[5, "price"] = 30.00
```

```
df.loc[30, "price"] = 15.00
```

```
df.loc[44, "price"] = 112.59
```

```
df.loc[49, "price" ] = 187.65
df.loc[50, "price" ] = 800.00
df.loc[55, "price" ] = 200.00
df.loc[91, "price" ] = 37.53
df.loc[103, "price" ] = 37.53
df.loc[127, "price" ] = 80.00
df.loc[176, "price" ] = 200.00
df.loc[245, "price" ] = 37.53
```

Step 9: By-hand check

In a markdown cell, compute the mean and median of [2, 4, 4, 6, 9] by hand, then verify both with Python. You should see: mean 5.0, median 4, in both places.

For the mean calculate the average by adding all the numbers and dividing by the count, for median sort the numbers and take the one in the middle.

The mean is $(2 + 4 + 4 + 6 + 9) / 5 = 5$

The median is 1=2, 2=4, 3=4, 4=6, 5=9, then after sorting, the central number is index 3=4.0

```
In [18]: my_list = [2, 4, 4, 6, 9]
df_2 = pd.DataFrame(my_list)
print("mean of my_list: ", df_2.mean())
print("median of my_list: ", df_2.median())
```

```
mean of my_list: 0    5.0
dtype: float64
median of my_list: 0    4.0
dtype: float64
```

Behzad Ghabaei - End Of Lab

```
In [ ]:
```